Nikhil Rajan

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EDUCATION

London School of Economics and Political Science — **London, UK**
MSc in Financial Statistics — *August 2026 - June 2027*

- **Coursework:** Stochastic Calculus, Mathematics of Market Microstructure, Fixed Income, Computational Methods in Finance, Derivatives, Machine Learning in Finance, Financial Statistics, Advanced Statistics.

Madras School of Economics — **Chennai, India**
Bachelor of Arts (Honours) in Economics — *August 2023 - June 2026*

- **Cumulative Grade Point Average:** 9.61/10.0 ⧫ **Gold Medalist (Rank 1/60)**
- **Dean's List (Distinction):** Semesters II - VI
- **Coursework:** Stochastic Processes, Advanced Time Series, Real Analysis, Advanced Differential Equations, Linear Algebra, Econometrics, Statistics, Game Theory, Money and Banking, Data Science, Python and MATLAB.

INTERNSHIP EXPERIENCE

Madras School of Economics — **Chennai, India**
Research Intern (Advisor: Dr. Naveen Srinivasan) — *December 2024 - January 2025*

- Investigated horizontal innovation and knowledge spillovers in the Romer growth model using analytical and numerical methods.
- Contributed to a new micro-founded model linking institutional quality, innovation, and long-run productivity.

V MACS Yosaney — **Chennai, India**
Finance Intern — *May 2024 - July 2024*

- Built an automated tool for generating P&L statements, balance sheets, and financial schedules, reducing reporting time and errors.
- Analyzed MSME financial statements and evaluated bank/NBFC credit products to understand documentation and risk requirements.

QUANTITATIVE RESEARCH PROJECTS

Liquidity Regimes as a Markov Chain and Optimal Execution as a Markov Decision Process *(Ongoing)*
Modelling liquidity as a four-state Markov process and exploring continuous-time optimal execution under persistent liquidity regimes.

Credit Rating Transitions and Markov Chains
Reconstructed Jarrow - Lando - Turnbull (1997) from first principles; derived hedging strategies and risk-neutral transition matrices.

Stochastic Welfare Analysis of Sin Taxation Under India's New GST Regime
Modelled cigarette taxation through a Markov-based behavioural transition model using NSSO microdata.

Hidden Markov Model Framework for Credit Stress Detection and Recession Early Warning *(Ongoing)*
Developing a hidden Markov and mean-reverting stochastic process to detect credit stress regimes, quantify their lead-lag relationship with recessions, and drive a backtested tactical allocation strategy.

Impact of MGNREGA on India's Labour and Development Outcomes: An Econometric Analysis
Used OLS models to analyse how MGNREGA spending shapes rural wages, labour intensity, and women's participation.

TEACHING ASSISTANTSHIPS

Foundations of Modern Finance I & II (Course Link) — **Sem IV**
Conducted quizzes and weekly assignments; explained CAPM, Stochastic Discount Factors, and derivatives.

Stochastic Processes — **Sem V**
Created detailed notes on martingales, Markov Decision Processes, and Brownian motion; led problem-solving sessions.

Time Series Analysis — **Sem VI**
Worked on weekly assignments, with the syllabus covering the first five chapters of Hamilton's Time Series Analysis.

Game Theory — **Sem VI**
Engaged with weekly assignments while working through core game theoretic concepts.

Multivariable Calculus — **Sem II, IV**
Delivered revision lectures on multivariate calculus; held tutorials guiding juniors through assignments and proofs.

Intermediate Microeconomics — **Sem V**
Designed assignments and grading rubrics on monopoly, exchange, and externalities.

EXTRACURRICULAR ACHIEVEMENTS

App Development - Eminence Class Recognition

- Developed an education-tech application under India's first project-based learning program by IIT Madras, Rhapsody Plus, Intellect, and UC San Diego.

Chess (selected):

- Placed 13^{th} on merit list in 1st SRM International Open FIDE Rapid Chess Tournament (2022)
- Captained the school team in CBSE Cluster Championships; scored 5.5/6 on Board 1 (2022)
- Placed 2^{nd} in Under-15 category in Talent Chess Academy's 2^{nd} FIDE-rated tournament (2019)

SKILLS

- **Quantitative & Financial Modeling:** Stochastic Processes, Econometrics, Time Series Analysis, Asset Pricing, Credit Risk Modeling, Game Theory.
- **Technical:** Python, MATLAB, Julia, Stata, Excel, Latex.
- **Analytical Foundations:** Real Analysis, Statistics, Differential Equations, Linear Algebra.